SDC

Hackathon · Phase 1

Preliminary Project Proposal

Lumina Nusantara · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

Submission deadline: 25 October 2026, 07:59:59 UTC+8

Phase 1 — Preliminary Project Proposal. Answers below follow the official template order and the stated word limits. Revised unsigned copy: leader review required.

1. TEAM AND PROJECT OVERVIEW

Team	Lumina Nusantara
Domain	Domain 1 — The Aiversecity Future-Ready Workforce Domain
SDG track	SDG 4 — Quality Education (sub-target 4.4, with 8.5)
Project	SkillRadar MY
Leader	DU YULIN
Contact	CYS2609009@xmu.edu.my / +60 11-1798 1756
University	Xiamen University Malaysia (XMUM)
Members	5: DU YULIN; GAO TANGLINYUE; ZHENG QUANYING; WU ZHUOLI; ZHANG BORAN

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1 Problem context

Malaysian workers and graduates need to understand how changing demand affects their next learning decision. The official SDC Domain 1 brief cites 56% AI-tool adoption versus 26% formal training and a MOSTI estimate of over 600,000 workers needing reskilling. These are secondary citations from the competition brief, not measurements collected by this team. Static course lists do not explain why a role fits, what skills are missing, or how the recommendation changes under uncertainty. We target workers, graduates, HR teams and career centres who need a transparent starting point for reskilling. The prototype supports guidance rather than predicting employment or making hiring decisions.

2.2 Proposed solution and current implementation

SkillRadar MY accepts a questionnaire or synthetic text CV. A local keyword and alias parser maps skills to a team-curated catalogue; it does not implement a complete O*NET/ESCO identifier mapping. Dated JSON snapshots separate DOSM occupation-level observations, JobStreet/Randstad named-skill signals and WEF global forecasts. Validated manual import supports updating selected snapshots; automatic collection is future work. Five visible scoring components cover skills, role focus, transferability, demand and gap cost. Ordered learning steps link to recorded resources or search routes where coverage is absent. Hours and fees still require verification. Role and skill SDG mappings are team interpretations. Moderate and fast-disruption scenarios expose sensitivity to assumptions. The current application uses Streamlit, Python, JSON and session state. It calls no external model API and requires no GPU. Replit usage screenshots are retained; Streamlit Community Cloud provides the public demo. Automated ingestion, semantic parsing and a database are future options, not implemented features.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1 Architecture

Synthetic profile / questionnaire -> Streamlit UI -> local skill extractor -> five-component matching engine -> learning pathway and SDG display. The engine reads team-curated JSON catalogues and dated trend snapshots. A human reviews source material and runs the validation/import command. There is no database or external inference service on the request path. Replit is the documented development workspace; the live URL uses Streamlit Community Cloud. No RunPod endpoint is used.

3.2 Design thinking and deployment

The problem definition draws on the Domain 1 brief and public labour-market reports. This revision does not claim completed interviews or user pilots: evidence for the previous five-person claims has not been supplied. The implemented flows expose missing skills, score components, evidence sources, scenarios and learning steps. Repository tests and development logs document local regression checks; these are not substitutes for

independent user validation. Before submission the team should check the deployed flow using synthetic data and restore judge access to the Replit project. A future pilot should observe users completing the same task with and without the prototype and record time, relevance ratings and confusion points. Deployment and human review must be confirmed by the team.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Targets: SDG 4.4 (employment-relevant skills), with 8.5 (decent work). Skill mappings express intended contribution, not measured UN impact.

Baseline: The brief reports 56% adoption and 26% training, a 30-percentage-point context gap. A product-specific time-to-plan baseline has not yet been measured. The previous 3-8 week / median four-week interview baseline is withdrawn pending evidence.

KPIs and proposed Phase 2 targets: Measure time to produce a sourced plan and participant-rated top-three relevance. Proposed pilot: at least 10 consenting adult testers using synthetic profiles; target median completion within five minutes and at least 70% rating one top-three role at least 4/5 for relevance. Target 20% lower median task time against an equivalent manual-search task, if a valid baseline is collected before the final. Audit 100% of displayed role/skill recommendations for a source or explicit team-assumption label. These are targets, not results. Longer-term learning completion is outside the final-round evaluation window.

5. ELIGIBILITY & READINESS CHECKLIST

SDG mapping and an implemented prototype are present. Team confirmation remains required for mandatory workshop attendance, official registration eligibility, deployed end-to-end testing, sponsor access, third-party terms and final submission completeness. Video is pending. This unsigned revision does not pre-check those attestations.

6. IP & ORIGINALITY DECLARATION

6.1 Prior-owned work: Previous team declaration: none. Leader must reconfirm before signing; repository history alone cannot establish originality.

6.2 Third-party materials: Python (PSF), Streamlit (Apache 2.0), pandas / NumPy (BSD 3-Clause), PyArrow (Apache 2.0), Matplotlib (Matplotlib license), ReportLab (BSD) and public-source references are used. The Technical Declaration inventories current data and development tools. Public availability is not a blanket reuse license; source-specific terms require team review.

6.3 Employment-related IP: Previous team declaration: none. Each member must reconfirm any employment, internship or consulting restrictions.

7. DECLARATION

By signing after review, the leader confirms the final submission is accurate and original to the best of the team's knowledge. This revised document has not been signed; the previous signature is not carried over to changed statements.

Leader signature (after review): _____

Name: DU YULIN

Actual signing date: _____